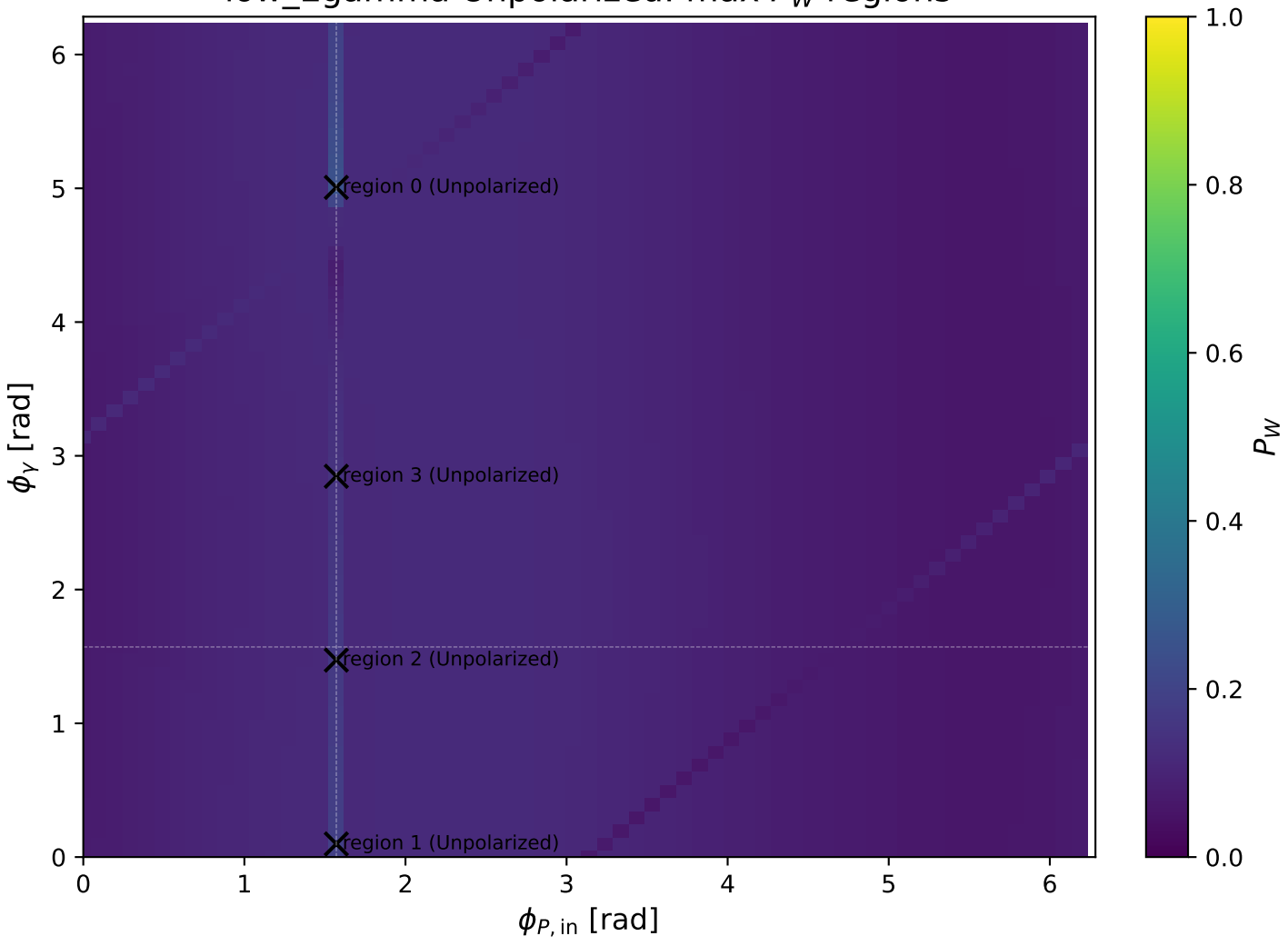


low\_Egamma Unpolarized: max  $P_W$  regions



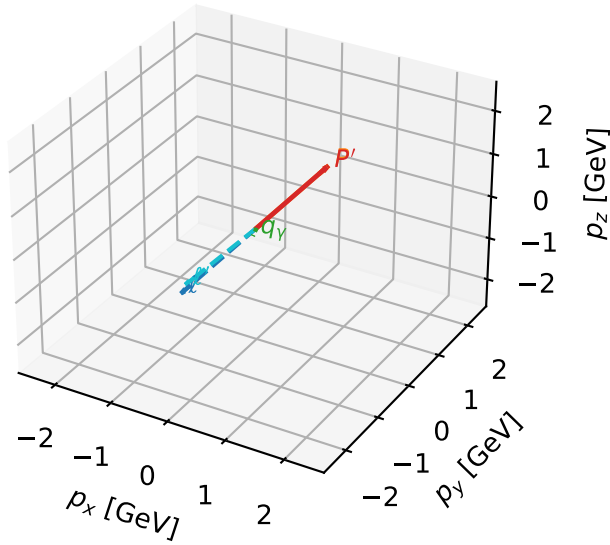
# Unpolarized characteristic max $P_W$ configuration

w\_purity\_low\_egamma\_unpolarized\_region\_0 P\_W=0.250894  
 region: low\_Egamma\_unpolarized\_region\_0  
 outgoing-state purity: 0.303092  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{4} \sum_{h_{\ell_0}, h_p} \rho(h_{\ell_0}, h_p)$

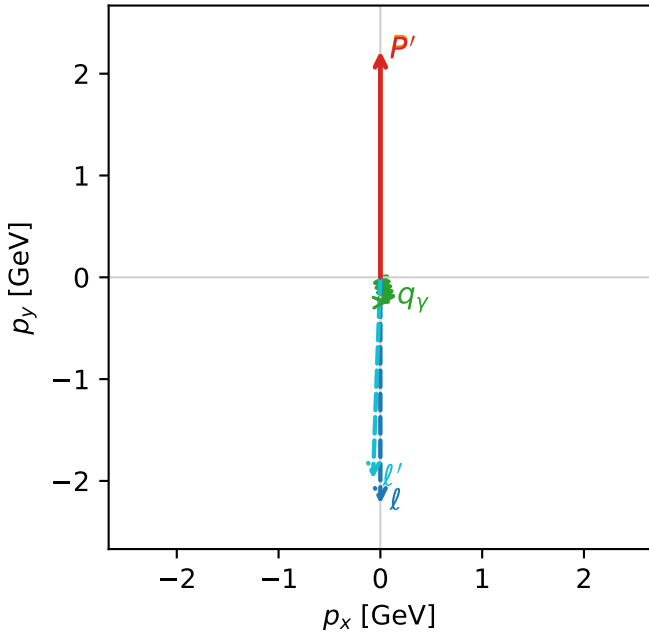
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.21812$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=4.71239$ ,  $\phi_p=1.5708$   
 $E_V=0.25$ ,  $\phi_V=5.00691$   
 $Q^2=0.00591804$ ,  $x_B=0.00260549$ ,  $t=-5.99798\text{e-}06$ ,  $W^2=3.1453$ ,  $y=0.110068$   
 $\theta(\ell', \gamma)=18.9753$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 1.9802$   $p_x= -0.0726$   $p_y= -1.9789$   $p_z= 0.0000$   
 $P'$   $E= 2.4083$   $p_x= 0.0000$   $p_y= 2.2181$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.0726$   $p_y= -0.2392$   $p_z= 0.0000$

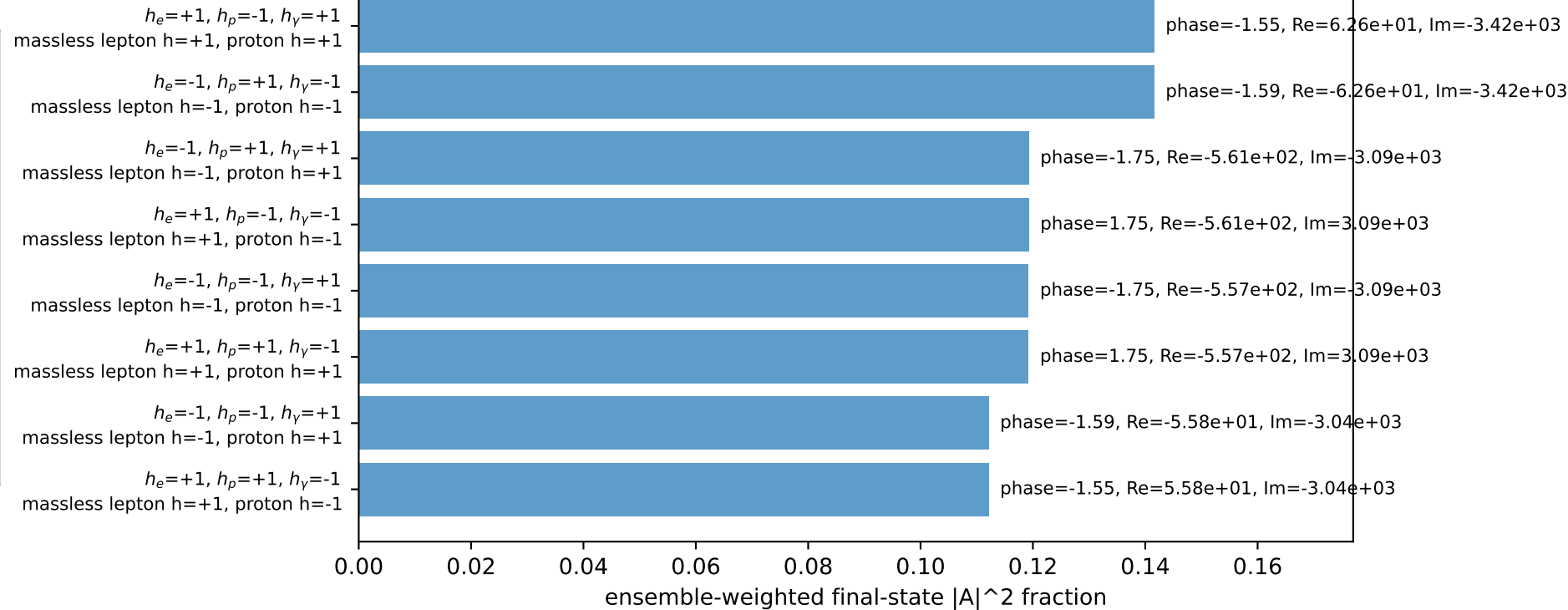
3D momenta



Transverse plane



Leading initial-ensemble/final-state components (N=8, retained=98.5%)



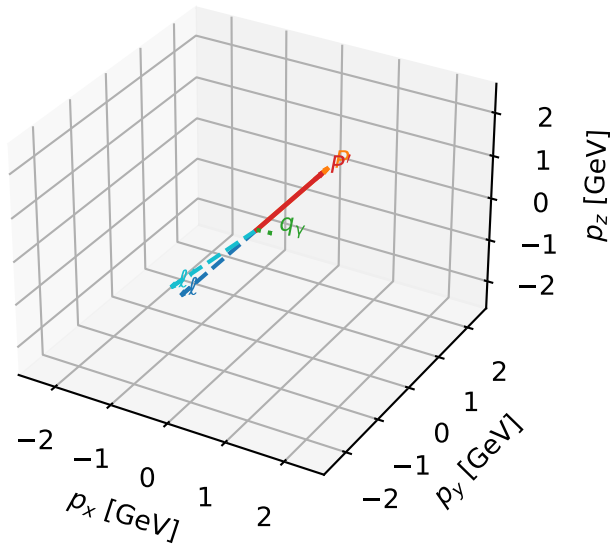
# Unpolarized characteristic max $P_W$ configuration

w\_purity\_low\_egamma\_unpolarized\_region\_1 P\_W=0.18909  
region: low\_Egamma\_unpolarized\_region\_1  
outgoing-state purity: 0.254781  
kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
incoming state:  $\frac{1}{4} \sum_{h_{\ell_0}, h_p} \rho(h_{\ell_0}, h_p)$

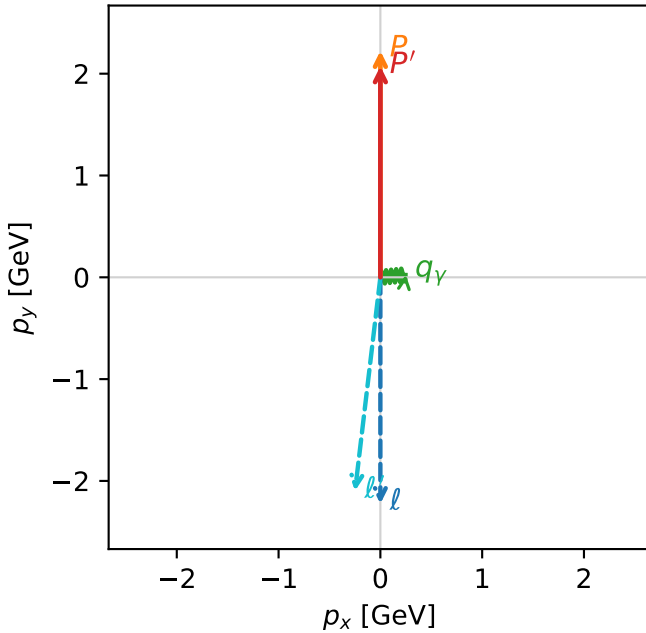
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.07351$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=4.71239$ ,  $\phi_P=1.5708$   
 $E_V=0.25$ ,  $\phi_V=0.0981748$   
 $Q^2=0.0653998$ ,  $x_B=0.0593636$ ,  $t=-0.00364768$ ,  $W^2=1.91613$ ,  $y=0.0533864$   
 $\theta(\ell', \gamma)=102.388$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1127$   $p_x= -0.2488$   $p_y= -2.0980$   $p_z= 0.0000$   
 $P'$   $E= 2.2758$   $p_x= 0.0000$   $p_y= 2.0735$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2488$   $p_y= 0.0245$   $p_z= 0.0000$

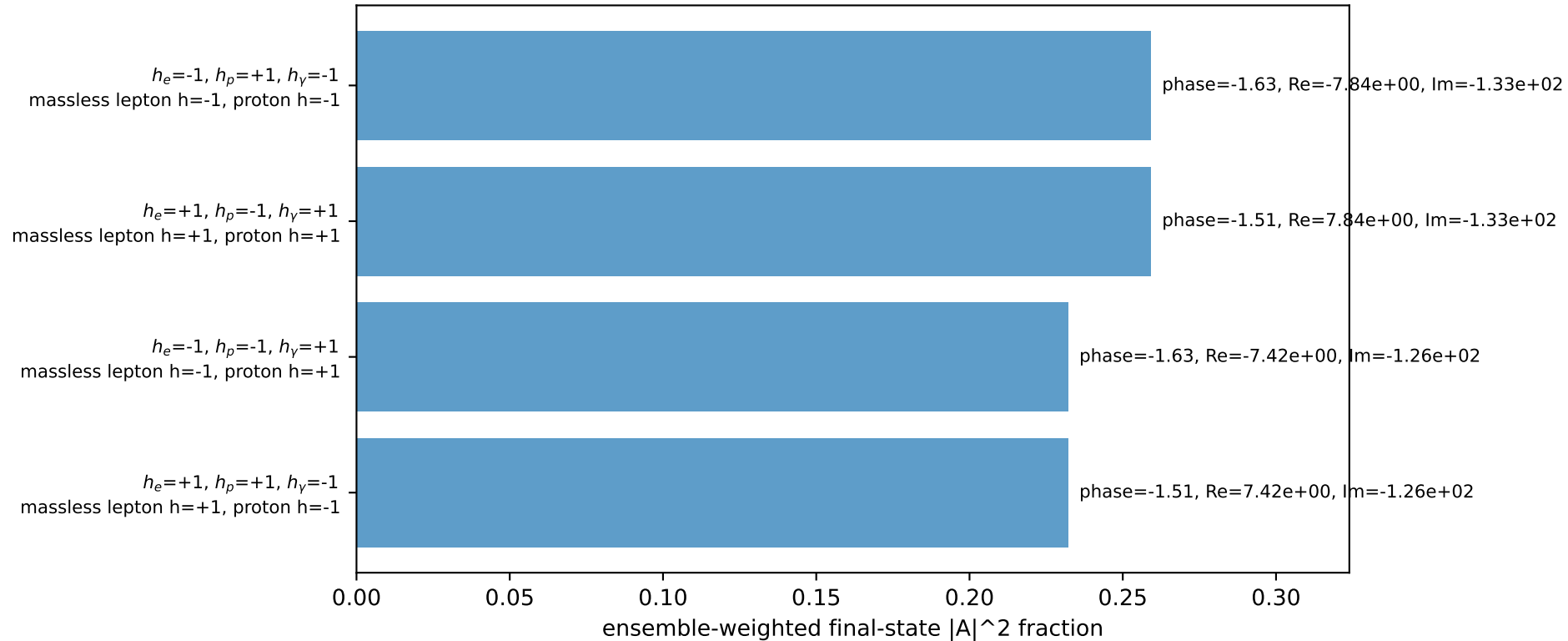
3D momenta



Transverse plane

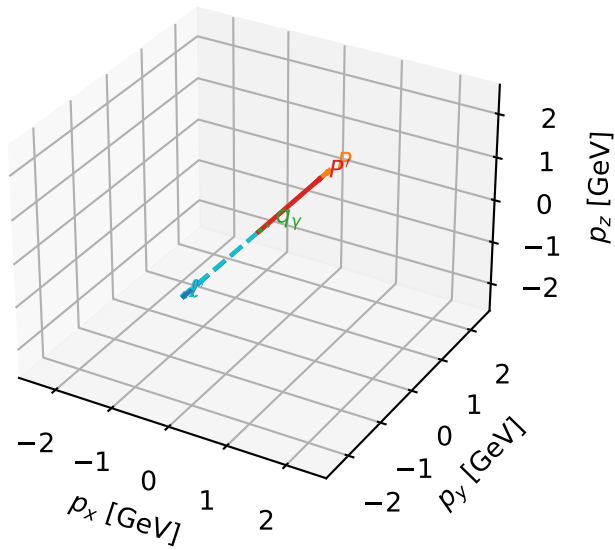


Leading initial-ensemble/final-state components (N=4, retained=98.3%)



# Unpolarized characteristic max $P_W$ configuration

3D momenta

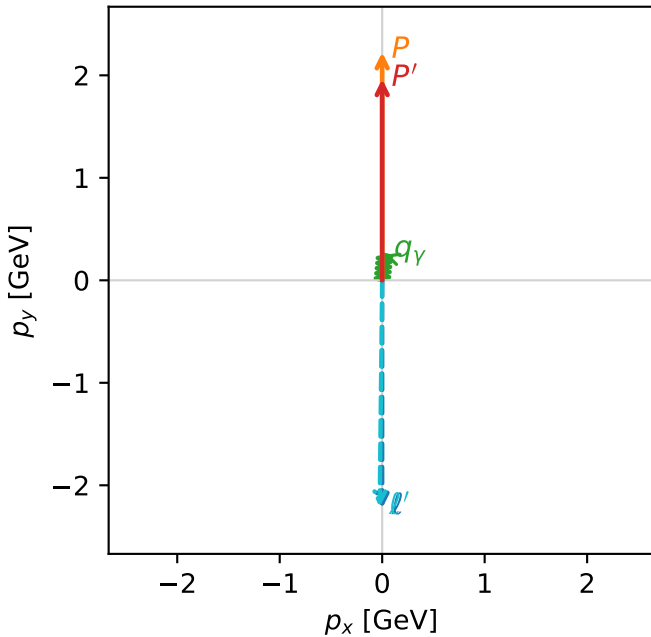


w\_purity\_low\_egamma\_unpolarized\_region\_2 P\_W=0.167907  
 region: low\_Egamma\_unpolarized\_region\_2  
 outgoing-state purity: 0.250021  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{4} \sum_{h_{\ell_0}, h_p} \rho(h_{\ell_0}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.96352$   
 $\theta_{in}=1.57079$ ,  $\phi_{\ell}=4.71239$ ,  $\phi_P=1.5708$   
 $E_Y=0.25$ ,  $\phi_Y=1.47262$   
 $Q^2=0.000603726$ ,  $x_B=0.00540958$ ,  $t=-0.0114065$ ,  $W^2=0.990843$ ,  $y=0.00540818$   
 $\theta(\ell', \gamma)=175.01$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.2125$   $p_x= -0.0245$   $p_y= -2.2123$   $p_z= 0.0000$   
 $P'$   $E= 2.1761$   $p_x= 0.0000$   $p_y= 1.9635$   $p_z= 0.0000$   
 $q_Y$   $E= 0.2500$   $p_x= 0.0245$   $p_y= 0.2488$   $p_z= 0.0000$

Transverse plane



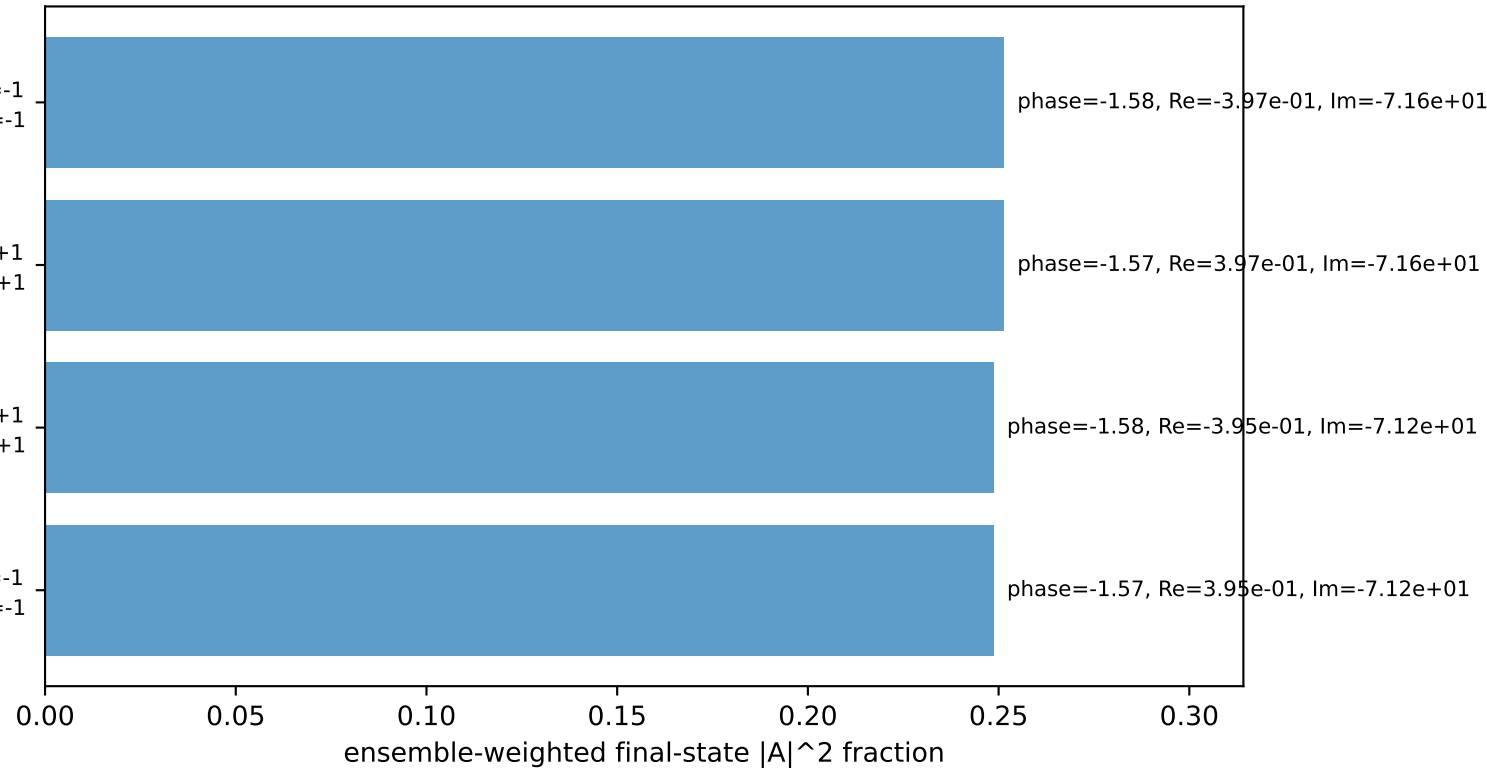
$h_e=-1, h_p=+1, h_Y=-1$   
 massless lepton  $h=-1$ , proton  $h=-1$

$h_e=+1, h_p=-1, h_Y=+1$   
 massless lepton  $h=+1$ , proton  $h=+1$

$h_e=-1, h_p=-1, h_Y=+1$   
 massless lepton  $h=-1$ , proton  $h=+1$

$h_e=+1, h_p=+1, h_Y=-1$   
 massless lepton  $h=+1$ , proton  $h=-1$

Leading initial-ensemble/final-state components (N=4, retained=100.0%)



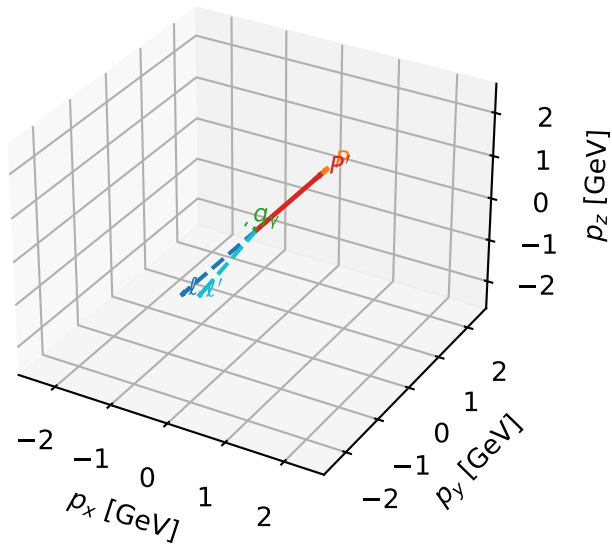
# Unpolarized characteristic max $P_W$ configuration

w\_purity\_low\_egamma\_unpolarized\_region\_3 P\_W=0.148199  
region: low\_Egamma\_unpolarized\_region\_3  
outgoing-state purity: 0.253285  
kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
incoming state:  $\frac{1}{4} \sum_{h_{\ell_0}, h_p} \rho(h_{\ell_0}, h_p)$

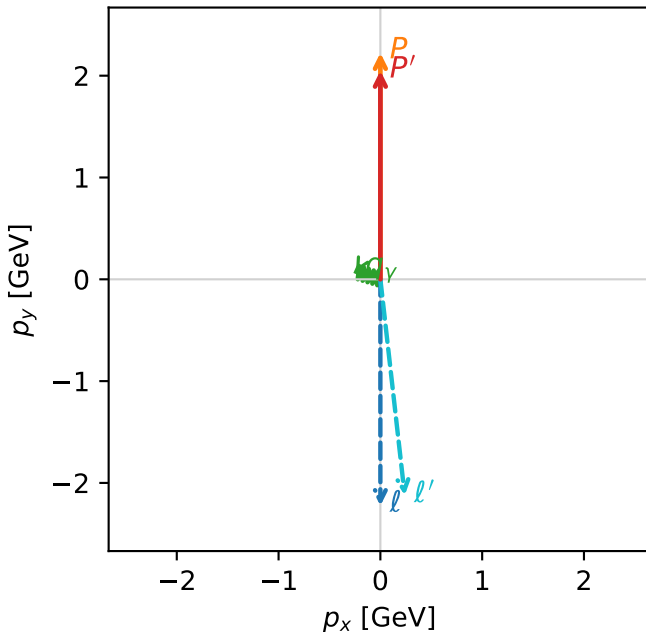
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.049$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=4.71239$ ,  $\phi_p=1.5708$   
 $E_V=0.25$ ,  $\phi_V=2.84707$   
 $Q^2=0.0598183$ ,  $x_B=0.0672743$ ,  $t=-0.00497769$ ,  $W^2=1.7092$ ,  $y=0.0430883$   
 $\theta(\ell', \gamma)=113.309$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1350$   $p_x= 0.2392$   $p_y= -2.1216$   $p_z= 0.0000$   
 $P'$   $E= 2.2535$   $p_x= 0.0000$   $p_y= 2.0490$   $p_z= 0.0000$   
 $q_V$   $E= 0.2500$   $p_x= -0.2392$   $p_y= 0.0726$   $p_z= 0.0000$

3D momenta



Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=98.8%)

